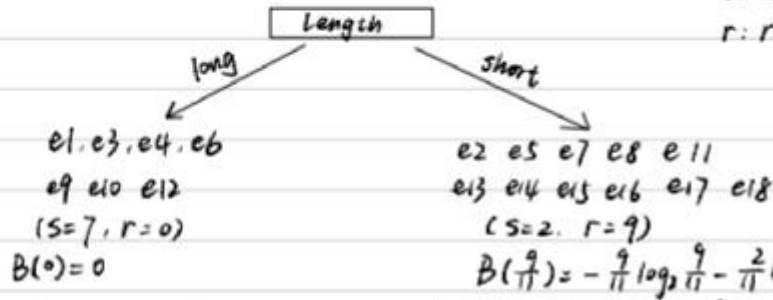


1.

$$B(\frac{9}{18}) = 1$$

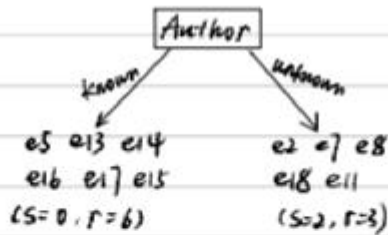
s: skips
r: reads

3.



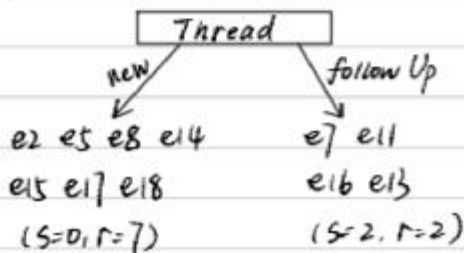
① Choose Author:

$$\text{Gain(Length)} = 1 - \frac{11}{18} B(\frac{9}{11}) \approx 0.5820$$



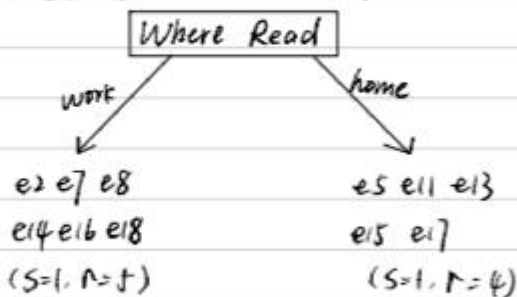
$$\text{Gain(Author)} = B(\frac{9}{11}) - \frac{6}{11} B(\frac{6}{6}) - \frac{5}{11} B(\frac{3}{5}) \approx 0.24270$$

② Choose Thread



$$\text{Gain(Thread)} = B(\frac{9}{11}) - \frac{7}{11} B(\frac{7}{7}) - \frac{4}{11} B(\frac{1}{2}) \approx 0.32040$$

③ Choose Where Read



$$\begin{aligned} \text{Gain(Where Read)} &= B(\frac{9}{11}) - \frac{6}{11} B(\frac{5}{6}) - \frac{3}{11} B(\frac{4}{5}) \\ &= 0.00133 \end{aligned}$$

故第 2 层选择 Thread

2.

解: (a)

A	B	C	D	E
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$(b) P(E|A,B,C,D) = \frac{P(A,B,C,D|E)P(E)}{P(A,B,C,D)} = \alpha P(E)P(A|E)P(B|E)P(C|E)P(D|E).$$

其中 $\alpha = \frac{1}{P(A,B,C,D)}$ 为正常数. 则:

$$\begin{aligned} P(e|a,b,c,d) &= \alpha P(e)P(a|e)P(b|e)P(c|e)P(d|e) \\ &= \alpha \times \frac{4}{15} \times \frac{2}{4} \times \frac{2}{4} \times \frac{2}{4} \times \frac{2}{4} \\ &= \frac{1}{60} \alpha \approx 0.0167\alpha \end{aligned}$$

$$\begin{aligned} P(\neg e|a,b,c,d) &= \alpha P(\neg e)P(a|\neg e)P(b|\neg e)P(c|\neg e)P(d|\neg e) \\ &= \alpha \times \frac{11}{15} \times \frac{5}{11} \times \frac{5}{11} \times \frac{5}{11} \times \frac{5}{11} \\ &= \frac{125}{3993} \alpha \approx 0.0313\alpha \end{aligned}$$

$$\therefore P(\neg e|a,b,c,d) > P(e|a,b,c,d)$$

$\therefore$ 预测此时 $E=0$.

感谢窦正同学的答案

3. 由 ML24-4 第 35 页和 36 页的推导, 有

$$\frac{\partial Loss_{k=k_0}}{\partial w_{j=j_0, k=k_0}} = -2(y_{k_0} - a_{k_0})g'(\sum_j w_{j, k_0} a_j) a_{j_0}$$

$$\frac{\partial Loss_{k=k_0}}{\partial w_{i=i_0, j=j_0}} = -2(y_{k_0} - a_{k_0})g'(\sum_j w_{j, k_0} a_j) w_{j_0, k_0} g'(\sum_i w_{i, j_0} a_i) a_{i_0}$$

其中, $in_j = \sum_i w_{i, j} a_i = \sum_i w_{i, j} x_i$, $in_k = \sum_j w_{j, k} a_j = \sum_k w_{j, k} g(in_j)$ 。

(a) 要计算 $\frac{\partial Loss_{o_1}}{\partial w_1}$, 即是要计算 $\frac{\partial Loss_{k=1}}{\partial w_{i=1, j=1}}$ 。从课件数据计算式中各项, 可得

- $y_1 = 0.01$, 输出层神经元 o_1 的输出 $a_{k=1} = 0.751365$;
- 输出层神经元 o_1 的输入 $in_{o_1} = \sum_j w_{j, k=1} a_j = 1.10596$, sigmoid 函数在 in_{o_1} 的导数 $g'(in_{o_1}) = \text{sigmoid}'(1.10596) = \text{sigmoid}(1.10596) \times (1 - \text{sigmoid}(1.10596)) = 0.186816$;
- $w_{j=1, k=1} = w_5 = 0.4$;
- 隐层神经元 h_1 的输入 $in_{h_1} = \sum_i w_{i, j=1} a_i = 0.3775$, sigmoid 函数在 in_{h_1} 的导数 $g'(in_{h_1}) = 0.2413$;
- 输入 i_1 的值 $a_{i=1} = 0.05$ 。

因此有

$$\frac{\partial Loss_{o_1}}{\partial w_1} = -2(0.01 - 0.751365) \times 0.186816 \times 0.4 \times 0.2413 \times 0.05 \approx 0.00133679$$

(b) 要计算 $\frac{\partial Loss_{o_2}}{\partial w_8}$, 即是要计算 $\frac{\partial Loss_{k=2}}{\partial w_{j=2, k=2}}$ 。从课件数据计算式中各项, 可得

- $y_2 = 0.99$, 输出层神经元 o_2 的输出 $a_{k=2} = 0.755523$;
- 输出层神经元 o_2 的输入 $in_{o_2} = \sum_j w_{j, k=2} a_j = 0.985700$, tanh 函数在 in_{o_2} 的导数 $g'(in_{o_2}) = 1 - g^2(in_{o_2}) = 1 - \tanh^2(0.9857) = 2\text{sigmoid}(1.9714) - 1 = 0.429186$;
- 隐层神经元 h_2 的输出 $a_{j=2} = g(\sum_i w_{i, j=2} a_i) = \tanh(0.25 \times 0.05 + 0.3 \times 0.1 + 0.35) = 2\text{sigmoid}(2 \times 0.3925) - 1 = 0.373513$ 。

因此有

$$\frac{\partial Loss_{o_2}}{\partial w_8} = -2(0.99 - 0.755523) \times 0.429186 \times 0.373513 \approx -0.0751764$$

4. Q4 感谢邓义圣同学的答案

解:(a)因为图像为 $5*5*3$,卷积核大小为 $3*3*3$,且卷积核只有一个,则 $(5-3)/2+1=2$,则输出大小为 $2*2*1$.设四个值为 a_0, a_1, a_2, a_3

$$a_0 = (1*1+1*0+2*(-1)+1*1+0*0+0*(-1)+0*1+1*0+0*(-1)) + (0*1+2*1+1*1+1*0+0*0+2*0+1*(-1)+0*(-1)+2*(-1)) + (1*1+1*0+0*1+1*0+1*(-1)+0*0+0*1+2*0+1*1)+1=2$$

对于 a_1, a_2, a_3 ,有类似的结果,由于权重共享,只需要将每个乘号前一项改为新的值即可.

$$a_1 = 0$$

$$a_2 = 6$$

$$a_3 = 3$$

故最终结果如下:

2	0
6	3

(b)在 $2*2$ 的最大池化中,输出结果为 $\max(a_0, a_1, a_2, a_3)$, a_0, a_1, a_2, a_3 为矩阵中的 4 个值.

对于最左上角的 $2*2$ 矩阵,输出为 $\max(6, 0, -7, 4)=6$

同理,对于剩下的输出结果依次为 16,10,15,14,14,4,19,0

6	16	10
15	14	14
4	19	0

(c)因为卷积核个数为 16,卷积核大小为 $6*6*3$,输入大小为 $100*100*3$,步长为 3,填充位为 1,则经过卷积层后输出大小为 $((100+1*2-6)/3+1)*((100+1*2-6)/3+1)*16$

即 $33*33*16$

经过池化层后 $W=(33-8)/5+1=6$

$$H=(33-8)/5+1=6$$

D 不变,仍为 16.

则池化层后输出为 $6*6*16$

5.

① 选择 healthy

$$Q[\text{healthy}, \text{party}] = 0.7 \cdot (10 + 0.9 \cdot 0) + 0.3 \cdot (10 + 0.9 \cdot 0) = 10$$

$$Q[\text{healthy}, \text{relax}] = 0.95 \cdot (7 + 0.9 \cdot 0) + 0.05 \cdot (7 + 0.9 \cdot 0) = 7$$

$$V[\text{healthy}] = \max(10, 7) = 10$$

Q[s,a]	party	relax
healthy	10	7
sick	0	0

	healthy	sick
V[s]	10	0

② 选择 sick

$$Q[\text{sick}, \text{party}] = 0.1 \cdot (2 + 0.9 \cdot 10) + 0.9 \cdot (2 + 0.9 \cdot 0) = 2.9$$

$$Q[\text{sick}, \text{relax}] = 0.5 \cdot (0 + 0.9 \cdot 10) + 0.5 \cdot (0 + 0.9 \cdot 0) = 4.5$$

$$V[\text{sick}] = \max(2.9, 4.5) = 4.5$$

Q[s,a]	party	relax
healthy	10	7
sick	2.9	4.5

	healthy	sick
V[s]	10	4.5

③ 选择 healthy

$$Q[\text{healthy}, \text{party}] = 0.7 \cdot (10 + 0.9 \cdot 10) + 0.3 \cdot (10 + 0.9 \cdot 4.5) = 17.515$$

$$Q[\text{healthy}, \text{relax}] = 0.95 \cdot (7 + 0.9 \cdot 10) + 0.05 \cdot (7 + 0.9 \cdot 4.5) = 15.7525$$

$$V[\text{healthy}] = \max(17.515, 15.7525) = 17.515$$

Q[s,a]	party	relax
healthy	17.515	15.7525
sick	2.9	4.5

	healthy	sick
V[s]	17.515	4.5

感谢吴莹菲同学的答案